

Day 21 (Part 2): Spanning Tree Protocol

STP STATES

STP Port State	Stable/Transitional
Blocking	Stable
Listening	Transitional
Learning	Transitional
Forwarding	Stable

- ROOT / DESIGNATED PORTS remain STABLE in a FORWARDING state
- NON-DESIGNATED PORTS remain STABLE in a BLOCKING state
- LISTENING and LEARNING are TRANSITIONAL states which are passed through when an interface is activated, or when a BLOCKING PORT must transition to a FORWARDING state due to a change in network topology.

1) BLOCKING / STABLE

- NON-DESIGNATED PORTS are in a BLOCKING state
- Interfaces in a BLOCKING state are effectively disabled to prevent loops
- Interfaces in a BLOCKING state do NOT Send/Receive regular network traffic
- Interfaces in a BLOCKING state do NOT forward STP BPDUs
- Interfaces in a BLOCKING state do NOT learn MAC ADDRESSES

2) LISTENING / TRANSITIONAL

- After the BLOCKING state, interfaces with the DESIGNATED or ROOT role enter the LISTENING state
- ONLY DESIGNATED or ROOT PORTS enter the LISTENING state (NON-DESIGNATED PORTS are ALWAYS BLOCKING)
- The LISTENING state is 15 seconds long by Default. This is determined by the FORWARD DELAY TIMER
- Interfaces in a LISTENING state do NOT Send / Receive regular network traffic
- Interfaces in a LISTENING state ONLY Forward/Receive STP BPDUs
- Interfaces in a LISTENING state does NOT learn MAC ADDRESSES from regular traffic that arrives on the interface

3) LEARNING / TRANSITIONAL

- After the LISTENING state, a DESIGNATED or ROOT port will enter the LEARNING state
- The LEARNING state is 15 seconds long by Default. This is determined by the FORWARD DELAY TIMER (same one used for both LISTENING and LEARNING states)
- Interfaces in a LEARNING state do NOT Send / Receive regular network traffic
- Interfaces in a LEARNING state ONLY Sends/Receives STP BPDUs
- Interfaces in a LEARNING state **learns** MAC ADDRESSES from regular traffic that arrives on the interface

1. FORWARDING / STABLE

- ROOT and DESIGNATED PORTS are in a FORWARDING state

- A PORT in the FORWARDING state operate as NORMAL
- A PORT in the FORWARDING state Sends/Receives regular network traffic
- A PORT in the FORWARDING state Sends/Receives STP BPDUs
- A PORT in the FORWARDING state **learns** MAC ADDRESSES

SUMMARY :

Spanning Tree Port States

STP Port State	Send/Receive BPDUs	Frame forwarding (regular traffic)	MAC address learning	Stable/ Transitional
Blocking	NO/YES	NO	NO	Stable
Listening	YES/YES	NO	NO	Transitional
Learning	YES/YES	NO	YES	Transitional
Forwarding	YES/YES	YES	YES	Stable
Disabled	NO/NO	NO	NO	Stable

STP TIMERS

Spanning Tree Timers

STP Timer	Purpose	Duration
Hello	How often the root bridge sends hello BPDUs	2sec
Forward delay	How long the switch will stay in the Listening and Learning states (each state is 15 seconds = total 30 seconds)	15sec
Max Age	How long an interface will wait <u>after ceasing to receive Hello BPDUs</u> to change the STP topology.	20sec (10* hello)

💡 SWITCHES do NOT forward the BPDUs out of their ROOT PORTS and NON-DESIGNATED PORTS - ONLY their DESIGNATED PORTS !!!

MAX AGE TIMER:

- If another BPDU is received BEFORE MAX AGE TIMER counts down to 0, the TIME will RESET to 20 Seconds and no changes will occur.
- If another BPDU is not received, the MAX AGE TIMER counts down to 0 and the SWITCH will re-evaluate it's STP choices, including ROOT BRIDGE, LOCAL ROOT, DESIGNATED, and NON-DESIGNATED PORTS.
- If a NON-DESIGNATED PORT is selected to become a DESIGNATED or ROOT PORT, it will transition from the BLOCKING state to the LISTENING state (15 Seconds), LEARNING state (15 Seconds), and then finally the FORWARDING state.
 - So... it can take 50 Seconds for a BLOCKING interface to transition to FORWARDING! (MAX AGE TIMER + (LISTENING + LEARNING 15 Second timers))
- These TIMERS and TRANSITIONAL STATES are to make sure that LOOPS are not accidentally created by an INTERFACE moving to FORWARDING STATE too soon

HOWEVER ...

💡 A FORWARDING interface can move DIRECTLY to a BLOCKING state (there is no worry about creating a loop)

💡 A BLOCKING interface can NOT move DIRECTLY to a FORWARDING state. It MUST go through the LISTENING and LEARNING states first!

STP BPDU (BRIDGE PROTOCOL DATA UNIT)

Ethernet Header of a BPDU

```
> Frame 999: 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface 0
> Ethernet II, Src: aa:aa:aa:aa:aa:ab (aa:aa:aa:aa:aa:ab), Dst: PVST+ (01:00:0c:cc:cc:cd)
> 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 10
> Logical-Link Control
▼ Spanning Tree Protocol
  Protocol Identifier: Spanning Tree Protocol (0x0000)
  Protocol Version Identifier: Spanning Tree (0)
  BPDU Type: Configuration (0x00)
  ▼ BPDU flags: 0x00
    0... .... = Topology Change Acknowledgment: No
    .... 0 = Topology Change: No
  ▼ Root Identifier: 32768 / 10 / aa:aa:aa:aa:aa:aa
    Root Bridge Priority: 32768
    Root Bridge System ID Extension: 10
    Root Bridge System ID: aa:aa:aa:aa:aa:aa (aa:aa:aa:aa:aa:aa)
    Root Path Cost: 0
  ▼ Bridge Identifier: 32768 / 10 / aa:aa:aa:aa:aa:aa
    Bridge Priority: 32768
    Bridge System ID Extension: 10
    Bridge System ID: aa:aa:aa:aa:aa:aa (aa:aa:aa:aa:aa:aa)
    Port identifier: 0x8002
    Message Age: 0
    Max Age: 20
    Hello Time: 2
    Forward Delay: 15
```

💡 PVST+ uses the MAC ADDRESS :

01 : 00 : 0c : cc : cc : cd

PVST = ONLY ISL Trunk Encapsulation

PVST+ = Supports 802.1Q

💡 Regular STP (not Cisco's PVST+) uses the MAC ADDRESS :

01 : 80 : c2 : 00 : 00 : 00

💡 The STP TIMERS on the ROOT BRIDGE determine ALL STP TIMERS for the entire network!

STP OPTIONAL FEATURES (STP TOOLKIT)

PORTFAST:

- Can be Enabled on INTERFACES which are connected to END HOSTS

💡 PORTFAST allows a PORT to move immediately to the FORWARDING state, bypassing LISTENING and LEARNING

- If used, it MUST be ENABLED only on PORTS connected to END HOSTS
- If ENABLED on a PORT connected to another SWITCH, it could cause a LAYER 2 LOOP

```
SW1(config)#interface g0/2
SW1(config-if)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
SW1(config-if)#
```

You can also ENABLE PORTFAST with the following command:

💡 SW1(config)# spanning-tree portfast default

This ENABLES PORTFAST on ALL ACCESS PORTS (not TRUNK PORTS)

BPDU GUARD:

- If an INTERFACE with BPDU GUARD ENABLED receives a BPDU from another SWITCH, the INTERFACE will be SHUT DOWN to prevent loops from forming.

```
SW1(config)#interface g0/2
SW1(config-if)#spanning-tree bpduguard enable
SW1(config-if)#
```

You can also ENABLE BPDU GUARD with the following command:

💡 SW1(config)# spanning-tree portfast bpduguard default

This ENABLES BPDU GUARD on all PORTFAST-enabled INTERFACES

ROOT GUARD / LOOP GUARD:

Root Guard	If you enable root guard on an interface, even if it receives a superior BPDU (lower bridge ID) on that interface, the switch will not accept the new switch as the root bridge. The interface will be disabled.
Loop Guard	If you enable loop guard on an interface, even if the interface stops receiving BPDUs, it will not start forwarding. The interface will be disabled.

You probably do NOT have to know these STP optional features (or others such as UplinkFast, Backbone Fast, etcetera) for the CCNA.

BUT...

💡 Make sure you know PORTFAST and BPDU GUARD.

STP CONFIGURATION

Command to CONFIGURE Spanning-Tree mode on a SWITCH

```
SW1(config)#spanning-tree mode ?
mst          Multiple spanning tree mode
pvst         Per-Vlan spanning tree mode
rapid-pvst   Per-Vlan rapid spanning tree mode

SW1(config)#spanning-tree mode pvst
```

Modern Cisco SWITCHES run **rapid-pvst**, by default

CONFIGURE THE PRIMARY ROOT BRIDGE

Command to CONFIGURE Spanning-Tree PRIMARY ROOT BRIDGE on a SWITCH

```
SW3(config)#spanning-tree vlan 1 root primary
SW3(config)#do show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     cccc.cccc.cccc
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577 (priority 24576 sys-id-ext 1)
             Address     cccc.cccc.cccc
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15 sec
```

Confirm with "(do) show spanning-tree"

Can see in the above example, SW3 has become the "root"

- The "spanning-tree vlan root primary" command sets the STP PRIORITY to 24576. If another SWITCH already has a priority number lower than 24576, it sets this SWITCH's priority to 4096 LESS THAN the other SWITCH's Priority (remember STP PART 1 lecture)

SECONDARY ROOT BRIDGE (backup ROOT BRIDGE)

Command to CONFIGURE Spanning-Tree SECONDARY ROOT BRIDGE on a SWITCH

```

SW2(config)#spanning-tree vlan 1 root secondary
SW2(config)#do show spanning-tree

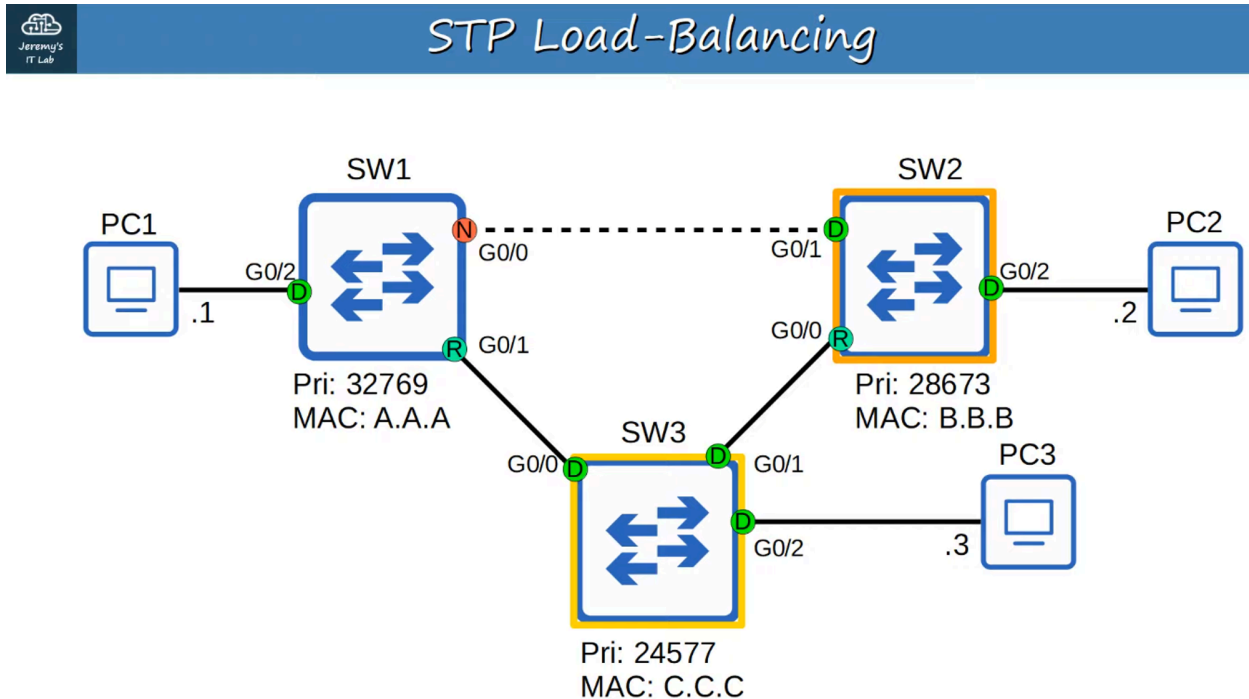
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     cccc.cccc.cccc
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28673 (priority 28672 sys-id-ext 1)
             Address     bbbb.bbbb.bbbb
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

```

- The "spanning-tree vlan root secondary" command sets the STP PRIORITY to 28672 (exactly 4096 higher than 24576).

VLAN 1 TOPOLOGY running PVST+

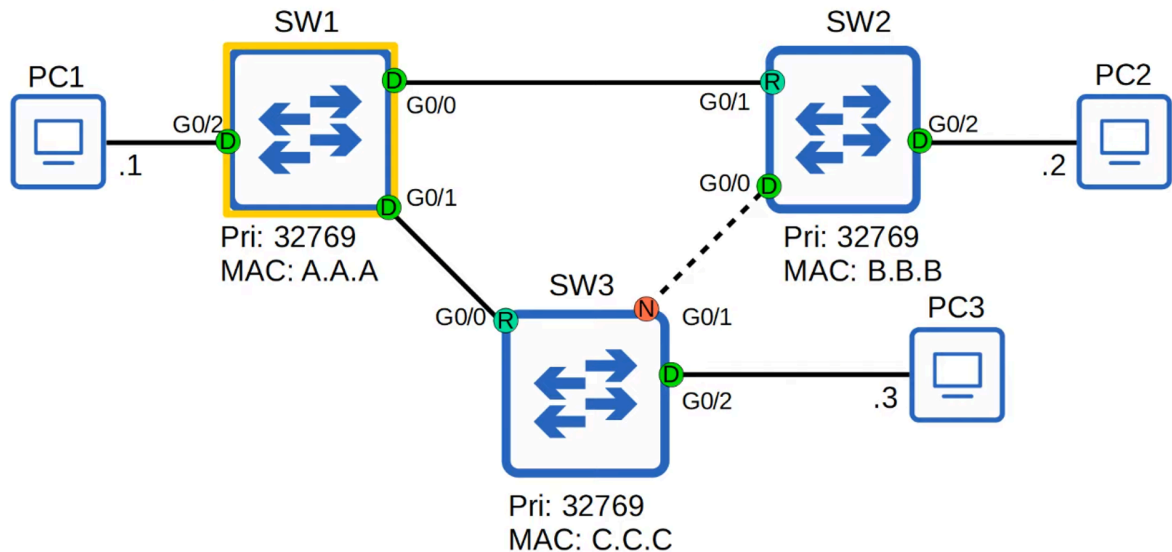


SW1 WAS the PRIMARY ROOT BRIDGE but :

- We have configured SW3 to be the PRIMARY
- We have configured SW2 to be the SECONDARY

The TOPOLOGY for VLAN 2, however, won't be the same. It will be the OLD Topology.

VLAN 2 Topology



WHY? Because we made changes ONLY to the TOPOLOGY found in VLAN 1 (see the commands we used)

CONFIGURE STP PORT SETTINGS

```

SW2(config-if)#spanning-tree vlan 1 ?
cost          Change an interface's per VLAN spanning tree path cost
port-priority Change an interface's spanning tree port priority

SW2(config-if)#spanning-tree vlan 1
  
```

"cost" = "ROOT COST"

"port-priority" = "PORT PRIORITY"